Explainable AI Code Generation Assistant

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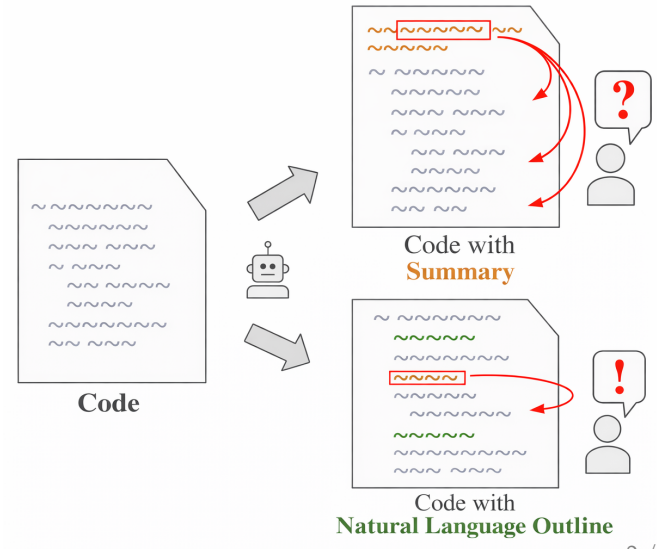
Problem

- LLMs write code, but do not explain decisions clearly.
- No mapping between summary and code.



Relevance

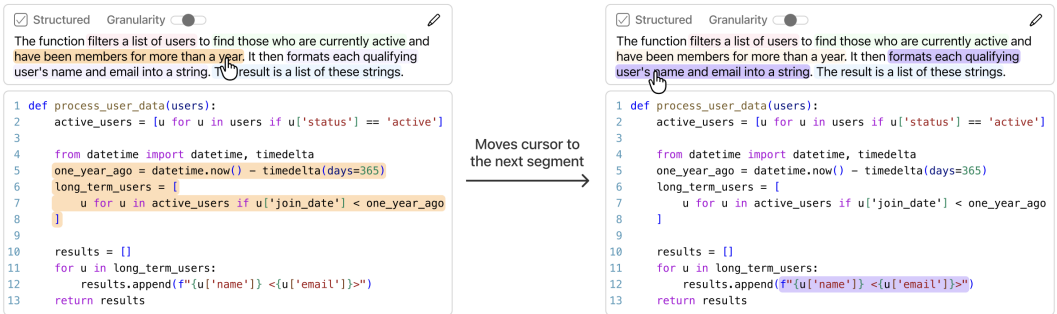
Researchers from Meta and academic collaborators introduced a framework for scaling test-time compute in long-horizon coding agents.

They transitioned away from using raw execution trajectories in favor of **structured summaries**.

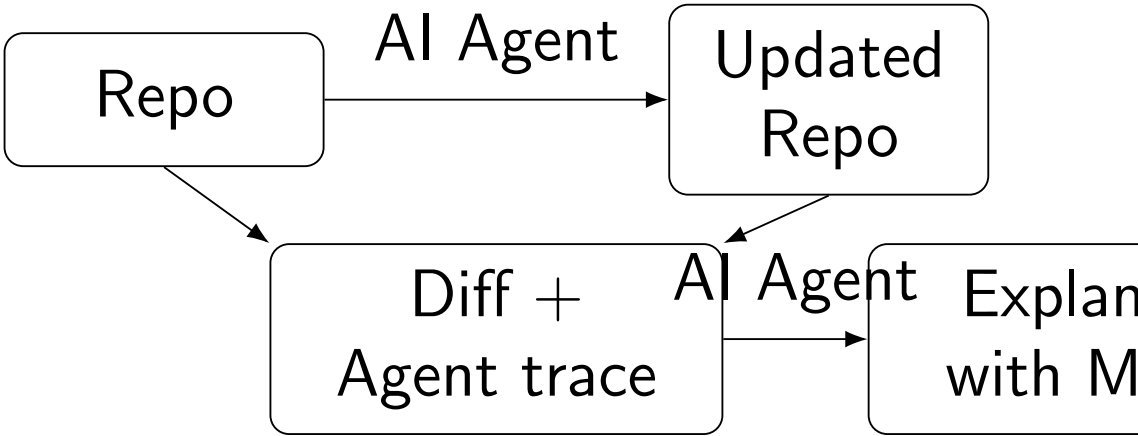
Scaling Test-Time Compute for Agentic Coding
arXiv, 16 Apr 2026
`https://arxiv.org/abs/2604.16529`

Goal

Build a code assistant that **generates code and human-friendly explanations** aligned with the code.



High-level architecture



Code with Explanations

Video demo is available in

`video_demo.mp4`

Open the exported HTML on GitHub Pages to play it inline.

SWE-bench Verified Mini: Django subset

What is in the dataset

- Real bug-fix tasks from `django/django`.
- For each task: GitHub issue text, starting commit, Django version, and setup commit.
- Reference solution: gold code patch and test patch.
- Evaluation tests: `FAIL_TO_PASS` checks fixed by the patch and `PASS_TO_PASS` regression checks.

Why it is needed

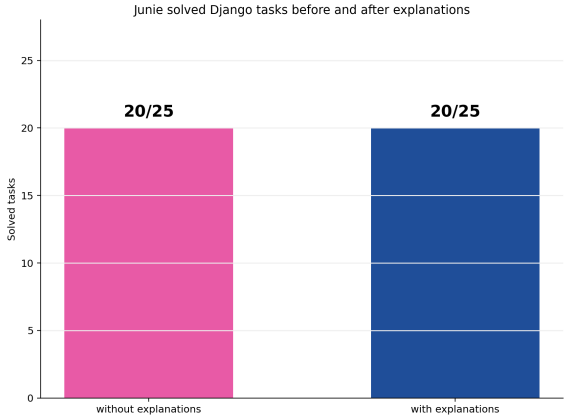
- Gives realistic repository-level tasks without running the full SWE-bench Verified set.
- Keeps evaluation focused on one codebase, so results are easier to compare and explain.
- Provides ground truth patches and tests for measuring whether generated code and explanations match the real fix.

Evaluation Plan

1. Run each Django task twice: baseline agent and agent with explanations.
2. Check correctness with the dataset test patch and existing regression tests.
3. Compare solved tasks: `PASS`/`FAIL` before and after explanations.
4. Compare agent efficiency: number of actions on tasks with passed evaluation.
5. Review explanations for grounding: do they describe the actual code changes?

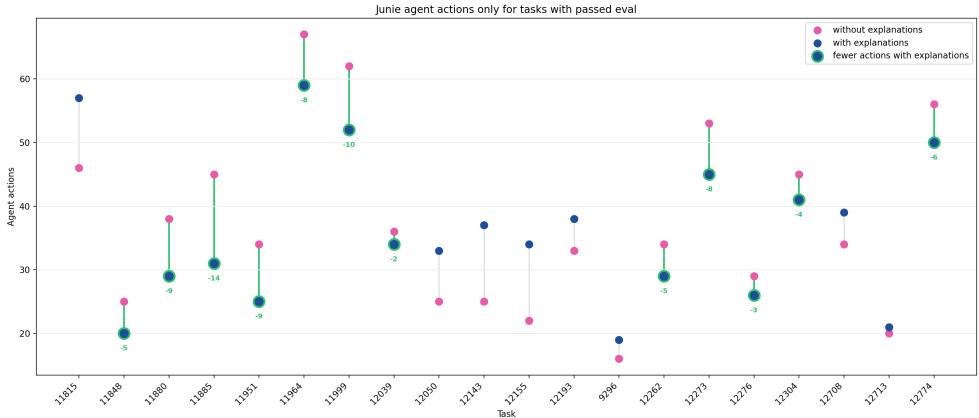
Explanations: Junie gemini-3.1-flash-lite-preview
Code generation: Junie gemini-3-flash-preview

Results



Solved Django tasks before and after adding explanations.

Results



Agent action counts on Django tasks with passed evaluation.

Summary

- Built a prototype for code generation with aligned explanations.
- Prepared a Django-only evaluation setup using SWE-bench Verified Mini.
- Initial result: solved tasks stayed at 20/25 after adding explanations.
- On passed tasks, explanations reduced Junie's actions in 12/20 cases.

Next: evaluate explanation grounding and extend experiments to more repositories and agents.